

SignalScope — one-page model report

SIH 2026 internal, Problem Statement 2 · model **mixed_resnet18_native_v1_calibrated** · SHA-256 eab7d9d82c8250bd... · 2026-09-12 · github.com/sibtainmunshi/Signal_Scope

Self-evaluated public-data results. Organizers supplied no dataset, baseline or hidden-test scores; those fields are **not supplied**, not estimated.

Task and scope

Binary real vs AI-generated image classification with a continuous AI-positive score; general objects, scenes and products only. Built: core detector, A (model-linked explanation + audit), C (robustness), D (EXIF only; C2PA not checked), F (local web app/CLI), G (bounded post-processing attack analysis).

Data and splits

Source	Use	Real / AI
CIFAKE (CIFAR-10 vs SD1.4, 32 px)	train 8,000 of 79,689	4,000 / 4,000
CIFAKE grouped val / calibration	selection / calibration	5000 / 4964 val
GenImage BigGAN + SD1.5 (train folders)	train 6,239	3,133 / 3,106
GenImage val / calibration	selection / calibration	369 / 414; 397 / 384
UniversalFakeDetect dev	evaluation only	guided vs ImageNet, LDM vs LAION, 500+500 each
Reserved final	scored once after freeze	GLIDE x3, DALLÉ vs LAION; CIFAKE author test 20k

Exact and perceptual-hash overlap exclusions against all protected data; grouped splits. Audit found real JPEG vs generated square PNG in both GenImage and the external set, so training balances JPEG and every external result is also reported **format-matched** (identical crop, resize and JPEG for both labels).

Model and training

- ImageNet-pretrained ResNet-18, fine-tuned end to end (AdamW 2e-4, cosine, 10 epochs, batch 64, AMP); epoch chosen by mean GenImage/CIFAKE val AUC.
- Native-resolution random 128 px crops (no resizing); every generated training image JPEG-compressed with qualities drawn from the real photos; symmetric rescale, blur, flip and crop-JPEG augmentation.
- Inference: mean logit of up to five native 128 px crops; CPU ~15–50 ms per image on our laptop.
- Temperature 1.65 fit on calibration splits; threshold 0.455 = strictest validation threshold with real FPR ≤ 5.0% per domain. ECE CIFAKE 0.027→0.009, GenImage 0.047→0.061 (worse).

Explanation (Module A)

Returned-class Grad-CAM per crop, stitched, plus a masking diagnostic. 40 fixed dev images (27 correct): masking the top Grad-CAM window lowered the verdict score more than random windows in 72.5% (mean fill, p=0.002) and 67.5% (blur); effects are small. JPEG map stability median rho 0.95; weight-randomization rho 0.52 (maps only partly weight-dependent). No named artifact (text, lighting, anatomy) is claimed.

Results at the frozen operating point (threshold 0.455)

Evaluation	ROC-AUC [95% CI]	Macro-F1	Accuracy	Real FPR	AI TPR
CIFAKE validation (in-domain)	0.993	0.951	95.2%	2.2%	92.5%
GenImage BigGAN/SD1.5 validation	0.956	0.898	89.8%	4.9%	85.0%
External dev, unseen generators	0.647 [0.624, 0.670]	0.513	57.8%	8.4%	23.9%
External dev, format-matched	0.653 [0.628, 0.677]	0.558	59.2%	13.9%	32.4%
Reserved GLIDE/DALLÉ (final)	0.565	0.400	42.1%	10.4%	18.3%
Reserved, format-matched (final)	0.643	0.472	47.9%	14.6%	29.2%
CIFAKE author test 20k (final)	0.993	0.951	95.1%	1.9%	92.2%

External per-generator AUC (as distributed; then matched): guided 0.630; ldm_200 0.665; guided 0.611; ldm_200 0.695. External F1, accuracy and rates are means over generator pairs.

GenImage validation confusion matrix

	Pred. real	Pred. AI
Real	351	18
AI	62	352

Baseline comparison (unseen-generator mean AUC, as distributed / matched)

Organizer baseline	not supplied
Our CIFAKE-only ResNet-18 (v0.1.0)	dev 0.553 / 0.540; reserved 0.548 / 0.546; dev real FPR 84.3%
Selected model	dev 0.647 / 0.653; reserved 0.565 / 0.643; dev real FPR 8.4%

Seven candidates were compared on the same development data (including a frozen CLIP head whose 0.708 as-distributed AUC fell to 0.542 when format-matched); reserved data were never used for selection.

Robustness (Module C/G)

GenImage val AUC original 0.956; JPEG q70 0.953; q30 0.906; half resolution 0.896; blur 0.937. Six-transformation search flipped 19.8% of initially correct predictions; on 32 px CIFAKE, half resolution drops AUC to 0.726 (flip rate 42.6%).

Limitations and failure cases

- Unseen-generator AUC is modest. On the reserved unseen set the mean AUC is 0.565 / 0.643 (as distributed / format-matched); GLIDE is weakest. At the conservative threshold most unseen AI images are missed.
- Pristine PNG outputs are harder than JPEG-compressed ones: training JPEG-balances generated images, so the model does not reward "no JPEG" as evidence of AI.
- Calibration is domain-specific: confidence and FPR need not transfer to new generators or pipelines.
- Resizing, blur and strong JPEG raise real-image false positives; tiny upscaled images are fragile.
- Grad-CAM shows model influence, not verified visual defects; audit sample is small and unannotated.
- GenImage-derived weights are for noncommercial research use (CC BY-NC-SA 4.0).